

# Enthalpy and Calorimetry

Exothermic and endothermic reactions,  $q = mc\Delta T$  and calorimetry

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

## USEFUL INFORMATION

$$q = mc\Delta T \cdot q_{\text{reaction}} = -q_{\text{solution}} \cdot \Delta H = q_{\text{reaction}}/n$$

$$c(\text{water}) = 4.184 \text{ J/(g}\cdot^{\circ}\text{C)} \cdot c(\text{copper}) = 0.385 \text{ J/(g}\cdot^{\circ}\text{C)} \cdot c(\text{iron}) = 0.449 \text{ J/(g}\cdot^{\circ}\text{C)}$$

Assume dilute solutions have density 1.00 g/mL and  $c = 4.184 \text{ J/(g}\cdot^{\circ}\text{C)}$  ·  $M(\text{CH}_4) = 16.04 \text{ g/mol}$ ,  $M(\text{NH}_4\text{NO}_3) = 80.04 \text{ g/mol}$ ,  $M(\text{C}_2\text{H}_5\text{OH}) = 46.07 \text{ g/mol}$

Show units in every step and state whether each  $\Delta H$  is exothermic or endothermic.

## A. Understanding

- Classify each process as exothermic or endothermic, and give the sign of  $\Delta H$ : (a) burning wood (b) melting ice (c) dissolving ammonium nitrate in water, which gets cold (d) neutralizing an acid with a base

- $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ ,  $\Delta H = -890.5 \text{ kJ}$ . (a) What is  $\Delta H$  for  $2\text{CH}_4 + 4\text{O}_2 \rightarrow 2\text{CO}_2 + 4\text{H}_2\text{O}$ ? (b) What is  $\Delta H$  for the reverse of the original reaction? (c) How much heat is released when 8.02 g of  $\text{CH}_4$  burns?

## B. $q = mc\Delta T$

- How much heat is needed to warm 40.0 g of copper from 25.0 °C to 75.0 °C?

4. A 100. g piece of metal absorbs 900. J and its temperature rises by 20.0 °C. Calculate its specific heat capacity.

5. 5.00 kJ of heat is added to 150. g of water at 22.0 °C. What is the final temperature?

### C. Calorimetry

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6. 8.00 g of ammonium nitrate is dissolved in 100.0 g of water in a calorimeter. The temperature falls from 22.0 °C to 16.0 °C. Calculate  $\Delta H$  for dissolving ammonium nitrate, in kJ/mol. (Mass of solution = 108.0 g.)

7. 25.0 mL of 2.00 M HCl and 25.0 mL of 2.00 M NaOH, both at 20.0 °C, are mixed. The temperature rises to 33.5 °C. Calculate  $\Delta H$  per mole of water formed.

### D. Challenge

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8. Burning 0.500 g of ethanol ( $\Delta H_c = -1366.8$  kJ/mol) under a can containing 200. g of water raises the water temperature by only 12.4 °C. (a) What temperature rise would occur if all the heat went into the water? (b) What percentage of the heat was transferred to the water?

9. A 50.0 g iron block at 100.0 °C is dropped into 100. g of water at 20.0 °C in an insulated cup. Find the final temperature. (Heat lost by the iron = heat gained by the water.)